



Society of Petroleum Engineers

SPE-227489-MS

Nanocellulose - A Multi-Functional Additive for Drilling Fluid Applications

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/227489-MS](https://doi.org/10.2118/227489-MS)

This paper was prepared for presentation at the Middle East Oil, Gas and Geosciences Show (MEOS GEO), Manama, Bahrain, 16 – 18 September 2025.

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Abstract

Global oil and gas industry has been drilling deeper, long horizontal, extended reach and multi-lateral wells in order to tap deeper reservoirs and to increase reservoir contact with the advent of technological breakthroughs as present hydrocarbon reservoirs are getting depleted. Exploration of new hydrocarbon reserves are generally found in challenging geological conditions which are associated with high pressure and high temperature environment. The conservative drilling fluid characteristically comprises of biopolymers such as cellulose, starch and/or xanthan gum for applications such as viscosifiers, filtration loss control additives and suspending agents. The working range of these additives are suitable for conventional drilling. Upon exposure to high pressure and high temperature environment drilling, these additives will undergo degradation and deterioration. Nano chemicals such as nanocellulose display outstanding thermal, physical and chemical properties owing to the well-ordered crystalline structure. Such property enhancements are essential to design a high performance water based drilling fluid systems. Regular acid-digestion procedure will remove the amorphous form of regular cellulose and retain only the nano-sized crystalline part of cellulose. Additional processing and chemical alteration of crystalline nanocellulose can yield to a variety of nano-based products such as polyanionic cellulose, hydroxyethyl cellulose, carboxymethyl cellulose, etc. for several oil and gas drilling applications. In this paper, we have explained the research and development efforts around nanocellulose for oil and gas industry drilling applications including techno-economic analysis of nanocellulose and the capability for oil and gas industry applications.

Introduction

The oil and gas process is dealing with different functional chemicals during the various stages of upstream operations such as drilling, cementing additives, completion and fracturing. A typical water-based mud is environment friendly and relatively less expensive as compared to oil and synthetic based mud. Water based drilling fluids perform during the majority of the operations. However, the additives in water based mud have limitations under harsh drilling. New oil and gas reserves discovered are mostly associated with high pressure and high temperature zones (Matthew 2004). Oil based drilling mud is also a choice to drill in the HPHT environment as it exhibits superior properties as compared to water based mud such as thermal stability, better lubricity, low formation damage, thin mud cake, resistance against microorganisms, etc.

Water based mud is a preferred choice as it is cheaper and environment friendly as compared to oil based mud (Amanullah et al. 2016). Synthetic mud on the other hand will also be stable under these conditions and is relatively non-toxic, but economically not favorable due to the cost. Hence, there is a requirement for a high performance water based mud system with additives with improved thermal, physical and chemical property. The advent of nanotechnology in the past years has made significant advancements in STEM research and unlocked a lot of avenues for new areas. Nanotechnology demonstrated that nanomaterials have superior chemical, physical, thermal and tribological properties than the corresponding macro and micro materials (Amanullah and Ramasamy 2014). Hence, the application of nanomaterials is very vital in designing a high performance water based drilling fluid for HPHT drilling applications.

A range of industries such as biomedical, coating, electrical, electronics, communications, etc. (Saidur et al. 2011) have already incorporated nanotechnology in the process very well and made substantial advancements in technologies over the past ten years. Cellulose based products have been a part of drilling fluid additives since the inception of drilling primarily owing to its accessibility and obtainability. Another advantage is the flexibility in handling as well as bio-degradable property. Though, cellulosic products offer notable fluid properties such as fluid loss control and rheology, the thermal stability is a fundamental limitation due to hydrolysis, oxidation and cleavage of chemical bonds at elevated temperatures. Poor salt tolerance and partial dispersibility in water are a few noted disadvantages. Many synthetic chemicals and polymers have been synthesized to overcome the above mentioned limitations. However, synthetic polymers do not have some of the crucial characteristics of biopolymers such as outstanding shear-thinning rheology. Therefore, an alteration in the process to make high performance cellulosic products that are able to withstand the harsh and challenging downhole conditions is a welcoming sign as this has inherent biodegradable property which makes it eco-friendly. These features suggests that the development of cellulosic nanomaterials for drilling fluid applications can provide high performance additives for extreme operating conditions. Compared to regular cellulose, nanocellulose exhibits excellent properties due to the distinctive morphologies, specific surface area, reactive hydroxyl groups on the surface, low coefficient of thermal expansion, high strength and modulus as well as gas permeability (Sales et al. 2014 and Spence et al. 2011). The ordered arrangement of the nanocellulose offer good thermal and chemical stability compared to conventional cellulose. The surface hydroxyl functional groups make them well dispersed in water and giving fluids with shear-thinning rheology and thixotropy and works well even at low concentrations. Additional advantage of reactive surface hydroxyl groups on nanocellulose includes further modification and functionalization possible (Hall 2017). Other general applications of nanocellulose are good fluid loss control, low friction and hence better rate of penetration, thin filter cake, therefore minimizing the chances of getting differential stuck, etc. Drilling fluids containing nano sized additives has received a lot of attention in the recent years (Ramasamy and Amanullah 2020, Vryzas and Kelessidis 2017). In this article, recent developments and applications of nanocellulose in drilling have been reviewed.

Nanocellulose

Cellulose is obtained from plants and woods and is one of the most abundant renewable bio-polymer resources available on earth with estimated production of around 10 bn tons per annum (Siqueira et al. 2010, Habibi 2010). Cellulose is a polysaccharide containing series of D-glucose units connected with each other as shown in Figure 1. Toughness, water insolubility, and fibrous properties play a vital role in the efficient properties of cellulose materials. Synthetic cellulose is also possible by algae, tunicate and bacteria (Rebouillat 2013). Other industries where cellulose is used include paper, food, cosmetics, pharmaceutical as well as in oil and gas industry for a range of applications. During the process of exploration and exploitation of hydrocarbon, cellulosic additives such as carboxymethyl cellulose (CMC) and polyanionic cellulose (PAC) have been regularly used in drilling fluids and fracturing fluids (Hughes et al. 1993). In the extreme HPHT drilling zones, the above-mentioned additives have limited performance because of inadequate physical, thermal, chemical and tribological stability (Thomas 1982).

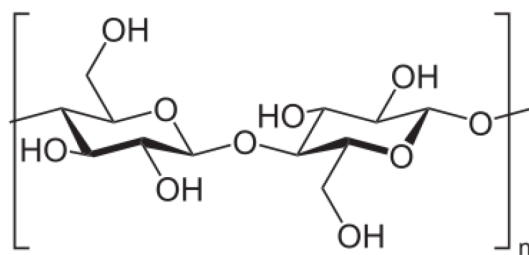


Figure 1—Structure of cellulose

Vital cellulosic characteristics such as biodegradability, optical, physical and mechanical properties can be significantly enhanced by changing to nano sized cellulose (Lee 2014). Many of the benefits of nanocellulose that are attributed to sustainability, mechanical properties such as large surface to volume ratio, high tensile strength and stiffness, high flexibility, and good electrical and thermal properties (Deepa et al. 2011). Enhancement of physical and chemical properties, such as high surface area (~ 250 m²/g), high tensile strength (7500 MPa), high stiffness (Young's modulus up to 140 GPa), and an abundance of surface hydroxyl groups make nanocellulose very attractive for different applications including biomedical engineering, waste water treatment, energy and electronics sector, etc. (Grishkewich et al. 2017). Dimensions of nanocellulose are in the range of 5–20 nanometers in width and up to 2000 nanometers in length.

Preparation Method

Cellulose is primarily produced for wood due to its availability and high cellulose content (Brinchi 2013). Based on the preparation method, different types of nanocellulose such as cellulose nanofibers (CNF), cellulose nanocrystals (CNC) and bacterial cellulose (BC) can be produced with different properties and dimensions. Bacterial Cellulose is produced by micro-organisms and have dimensions ranging from 6-50 nm width and a few micro meters in length (Moon 2011, Dufresne 2012). Cellulose nanofibers can be generated using mechanical treatment that contains both crystalline and amorphous forms having 5-60 nm width and 500 nm to 100 micro meter in length (Moon 2011; Klemm 2011; George and Montemagno 2017). Acid hydrolysis is the ideal choice of preparation method for highly crystalline, rigid nanocrystalline cellulose (CNC) with dimensions ranging from 3-5 nm in width and 50-500 nm in length (Habibi 2010; Moon 2011). Amorphous part of cellulosic fibers is more prone to acid attack whereas crystalline regions are more stable. Therefore, acid hydrolysis breaks out the disordered, irregular amorphous portion while leaving the more rigid and acid resistant nanocrystalline part unchanged.

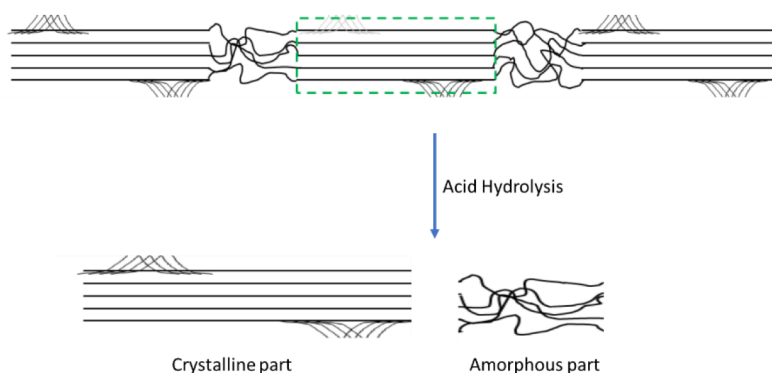


Figure 2—Representation of cellulose structure with crystalline and amorphous forms

The type of acid chosen for hydrolysis has an impact on the dimensions, stability and yield of CNC. The choice of sulfuric acid over hydrochloric acid will maximize the surface charges (sulfates groups) on CNC.

These charged functional groups increase water dispersibility and lead to stable colloidal suspensions in water (Bondeson et al. 2008, [Szczesna-Antczak et al. 2012](#)). On the other hand, the disadvantage of the ester groups on surface will lead to poor thermal stability and rheological behavior ([Habibi et al. 2010](#)). Different functionalities such as sulfate, hydroxyl, phosphate, carboxylate and amine functionalized CNC could be produced by selecting suitable hydrolysis and isolation process as described by [Börjesson et al. \(2015\)](#). The desired surface charges such as neutral, cationic and anionic can also be obtained by choosing appropriate process.

CNC functionalized with quaternary amine is prepared by reacting CNC slurry with 2,3-epoxypropyl trimethyl ammonium chloride and sodium hydroxide solution in ethanol for 2 h at elevated temperature as reported by Hauser and Tabb (2001). Oxidation of CNC is performed using 2,2,6,6-Tetramethylpiperidin-1-yl)oxyl (TEMPO), sodium bromide and sodium chlorate as reported by Saito et al (2006). CMC version of nanocellulose could be prepared by reacting CNC with 2-propanol, sodium hydroxide and chloroacetic acid as reported by Heinze (1994). All the three reaction schemes for the preparation of quaternary functionalized CNC, oxidized CNC and CMC-CNC are given below.

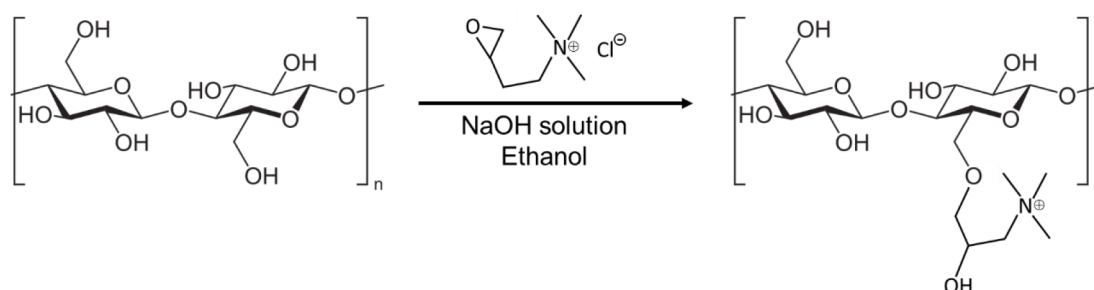


Figure 3—Reaction scheme to get quaternary ammonium functionalized nanocellulose

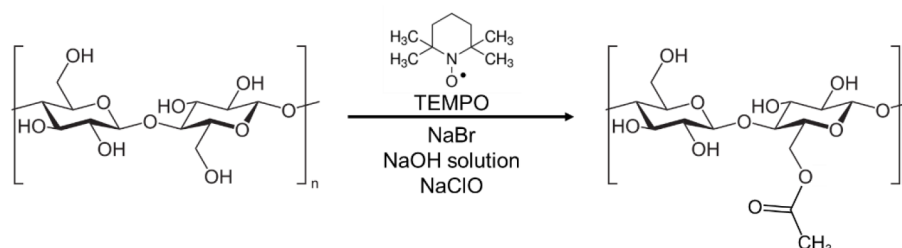


Figure 4—Reaction scheme for TEMPO oxidation of nanocellulose

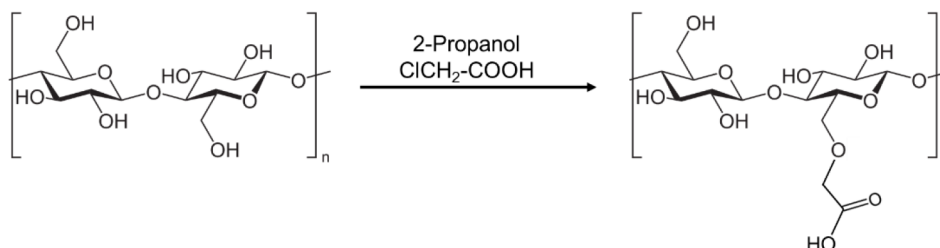


Figure 5—Reaction scheme for carboxylation of nanocellulose

Oil and gas industry applications of nanocellulose

Several bio-based additives have been used in oil and gas industry during various operational stages such as drilling, cementing and fracturing operations functioning as fluid loss control additive and viscosifying agent which govern technical performance of drilling fluid required for safe and economic drilling operation. Poor maintenance of these two properties will lead to a lot of different drilling issues such as excessive fluid

invasion to the formation causing damage to formation which will be an issue during production, thicker mud cake deposition on the borehole wall which may lead to stuck pipe problem, poor cuttings lifting and carrying ability which will lead to inefficient hole cleaning, etc. As per API recommendation, fluid loss has to be below certain value is crucial to maintain the designated drilling fluid viscosity, borehole formation stability, avoiding damage in pay zone, etc. High performance fluid loss additives are vital to minimize damage to formation especially while drilling the pay zone as mud particles in the filtrate will plug the pores leading to reduction in permeability (Mungun 1965). Todd et al. (1990) reported that water suspension having particle concentration as low as 1-15 ppm and of size smaller than 3 micron may cause significant damage to oil and gas reservoirs. Additives capable of creating thin and low permeable filter cake as soon as contacting the formation of borehole will reduce the near wellbore formation damage while drilling. Another major drilling issue encountered due to thick and improper formation of mud cake especially across high permeable zone will lead to stuck pipe which is one of the major drilling problems faced by oil and gas industry. It is reported that occurrence of stuck pipe is significantly higher while drilling with water-based muds than with the oil-based muds as the later have thin mud cake deposition (Chilingarian and Vorabutr 1983). Hence, mud additives that ensure the deposition of a very thin filter cake while drilling with water-based fluid will decrease the frequency of pipe sticking problems in differential sticking prone zones. The pulling force required for releasing a stuck pipe also increases with increasing mud cake thickness (Outmans 1958). Isambourg et al showed the magnitude of pulling force necessary to release a stuck pipe related to the thickness of the filter cake deposited on the formation (Isambourg and Marti 1999). Thick filter cake also has impact on casing placement especially in deviated, horizontal and extended reach wells due to the requirement of higher casing running load. A thin, low friction and slippery mud cake is very important for these wells (Maidla 1987). Moreover, a thick, loosely packed mud cake can result to high torque and drag in a long horizontal well when compared to an ultrathin, tightly packed mud cake.

Application of nanocellulose in drilling fluids

Several studies have been carried out in the recent past for evaluating the application of nanocellulose as fluid loss control additive to obtain lower filtration and to have thin mud cake. The effect of nano sized carboxymethyl cellulose and polyanionic cellulose as fluid loss additive have been studied by Saboori et al. (2102) and Fereydooni et al. (2012). The authors noted that a desirable reduction in filtration loss and cake thickness in API fluid loss experiments. Lee et al. have reported the formulation of high performance water based drilling fluid using nanocellulose and its derivatives (Hall et al. 2017). They have measured regular fluid properties of drilling fluids formulated using different derivatives of nanocellulose namely unmodified nanocellulose (CNF-0), carboxymethylated nanocellulose (C-CNF) and TEMPO-oxidized nanocellulose (TO-CNF) and the cationic cellulose with quaternary amine group (Cat-CNF). It is worth stating that the cationic quaternary ammonium variants of CNC are known to be very effective as shale inhibitors and clay stabilizers (Arvind 2009). Shale formation is usually drilled with highly inhibitive water based mud to prevent swelling and collapsing of clay particle upon contact with water. The Cat-CNF with quaternary ammonium compounds replace small cations like Na^+ and stabilize the shale by reducing the water invasion into shale layers and prevent it from swelling. The thermal stability of original cellulose nanoparticle (CNF) was compared against xanthan gum (XG) which is a routinely used viscosifier in water based drilling fluids using dynamic aging experiments carried out at elevated temperature of 300 °F under 125 psi pressure. The comparison of rheological property behavior of the muds with different CNCs is shown in figure 6. It is obvious that the rheological behavior of fluid having CNF-0 does not change much before and after subjecting to dynamic aging. On the other hand, the effect of temperature is obvious in the case of mud having xanthan gum as viscosifier. The high thermal stability of CNC derivatives keeps the fluid system intact even at higher temperature as compared to xanthan gum which degrades under high temperature conditions, as a result losing its viscosifying ability.

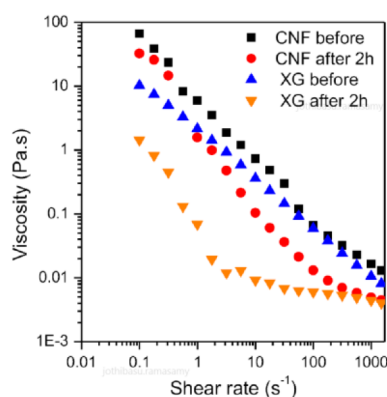


Figure 6—Viscosity profile of 0.5 wt% of suspension of CNF and XN (Hall et al. 2017).

Li et al. (2015) reported another application of cellulose nanocrystals as rheology modifier in figure 7. CNC used in this study has been obtained by hydrolysis of micro fibrillated cellulose (MFC) using sulfuric acid. The authors proved the efficiency of cellulose nanocrystals in bentonite mud with higher rheological properties, better thermal stability, minimal fluid loss and thinner mud cake deposition. The bentonite mud containing CNC has enhanced rheological and fluid loss properties than the bentonite mud having MFC. Moreover, the addition of polyanionic cellulose to bentonite mud having nanocellulose imparted a synergistic effect with nanocellulose which is reflected in additional improvement in rheological and fluid loss control performance. The API fluid loss of mud with CNC shows around 50% reduction in filtrate volume as compared to the mud with no nanocellulose. Moreover, mud with nanocellulose generated more compact and uniform filter cake. This highlights the potential of nanocellulose materials in overcoming the current and future drilling challenges.

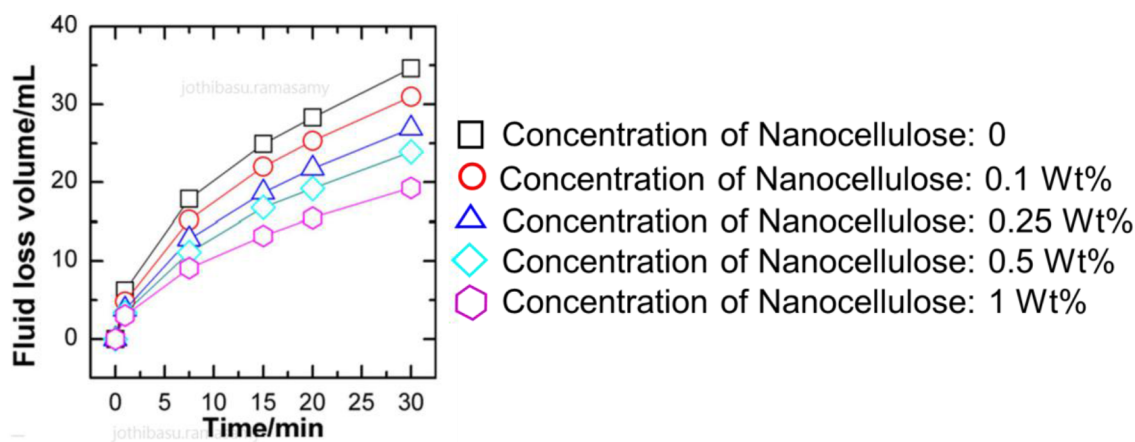


Figure 7—API fluid loss behavior of bentonite mud with different nanocellulose concentrations (Li et al. 2015).

Recommendations

Cellulose nanocrystals and cellulose nanofibers have physical, mechanical and chemical properties as compared to regular cellulose. Combination of these properties with other unique features of nanocellulose such as sustainability, bio-degradability and environment friendly nature make nanocellulose the ideal choice for applications in oil and gas industry especially in off-shore, deep water, extreme high pressure and high temperature environment drilling. The details available in the literature demonstrates that the application of nanocellulose in drilling fluids in order to improve mud properties such as viscous characteristics, enhanced fluid loss behavior and generation of thin mud cake. The source is abundant in nature and the results of the preliminary studies are very promising that nanocellulose based products

are indeed a good candidate for drilling fluids. The development, functionalization and application of nanocellulose is in the nascent stage and the full potential is yet to be unearthed with respect to oil and gas industry by conducting innovative research to overcome technical as well as environmental limitations.

Acknowledgement

The authors thank EXPEC Advanced Research Center of Saudi Aramco for the support and encouragement for this publication.

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